

Question 1

P2

Avery has 4 adventure books, 5 mystery books, and 1 comic book.

Determine the number of ways he can arrange all of the books on his shelf if each type of book must be grouped together.

Solution

$$\frac{3!}{\text{types of books}} \cdot \frac{4!}{\text{adventure books}} \cdot \frac{5!}{\text{mystery books}} \cdot \frac{1!}{\text{comic book}} = 17280 \text{ ways}$$

1 mark for arrangement of types of books
1 mark for arrangement of adventure, mystery, and comic books

2 marks

Note:

- 1! does not need to be shown.

Question 2

T5, T6

Solve the following equation, algebraically, over the interval $[0, 2\pi]$.

$$5 \cos^2 \theta - \cos \theta - \sin^2 \theta = 0$$

Solution

$$5 \cos^2 \theta - \cos \theta - (1 - \cos^2 \theta) = 0$$

1 mark for substitution of an appropriate identity

$$5 \cos^2 \theta - \cos \theta - 1 + \cos^2 \theta = 0$$

$$6 \cos^2 \theta - \cos \theta - 1 = 0$$

$$(3 \cos \theta + 1)(2 \cos \theta - 1) = 0$$

$$\cos \theta = -\frac{1}{3}$$

$$\cos \theta = \frac{1}{2}$$

1 mark for solving for $\cos \theta$ ($\frac{1}{2}$ mark for each branch)

$$\theta_r = 1.230\ 959$$

$$\theta = 1.911, 4.373$$

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

2 marks for solving for θ ($\frac{1}{2}$ mark for each value of θ)

or

$$\theta = 1.911, 4.373$$

$$\theta = 1.047, 5.236$$

4 marks

Given that $-6048x^2y^5$ is the sixth term in the expansion of $(3x - m)^7$, determine m .

Solution

$$-6048x^2y^5 = {}_7C_5(3x)^2(-m)^5 \quad 2 \text{ marks (1 mark for } {}_7C_5; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$-6048x^2y^5 = 21(9x^2)(-m^5)$$

$$-6048x^2y^5 = -189x^2m^5$$

$$32y^5 = m^5$$

$\frac{1}{2}$ mark for simplification

$$2y = m$$

$\frac{1}{2}$ mark for m

3 marks

Question 4

R10

A series of blood tests measures the concentration of a prescribed drug. This concentration decreases according to the formula $A = Pe^{rt}$ where:

A is the concentration at time t

P is the initial concentration

r is the rate of change

t is the time, in hours, after the first blood test

The initial concentration is 3.8900 units/mL. Three hours later, the concentration is 1.7505 units/mL.

- a) Determine the rate of change, r , algebraically.
- b) Determine the concentration of the prescribed drug four hours after the initial concentration of 3.8900 units/mL was measured. Express the answer correct to 4 decimal places.

Solution

a) $1.7505 = 3.8900e^{r(3)}$ $\frac{1}{2}$ mark for substitution

$$0.45 = e^{3r}$$

$$\ln 0.45 = 3r \ln e$$

$$\frac{\ln 0.45}{3} = r$$

or

$$-0.266\ 169\dots = r$$

2 marks

b) $A = 3.8900e^{-0.266\ 169(4)}$

$$A = 1.341\ 424\dots$$

$$A = 1.3414 \text{ units/mL}$$

1 mark for answer consistent with a)

1 mark

Question 5

T1

Ariane uses the formula $s = \theta r$ to determine the arc length of a circle that has a central angle of 20° and a radius of 15 cm.

Below is Ariane's work:

$$\begin{aligned} s &= \theta r \\ s &= (20)(15) \\ s &= 300 \text{ cm} \end{aligned}$$

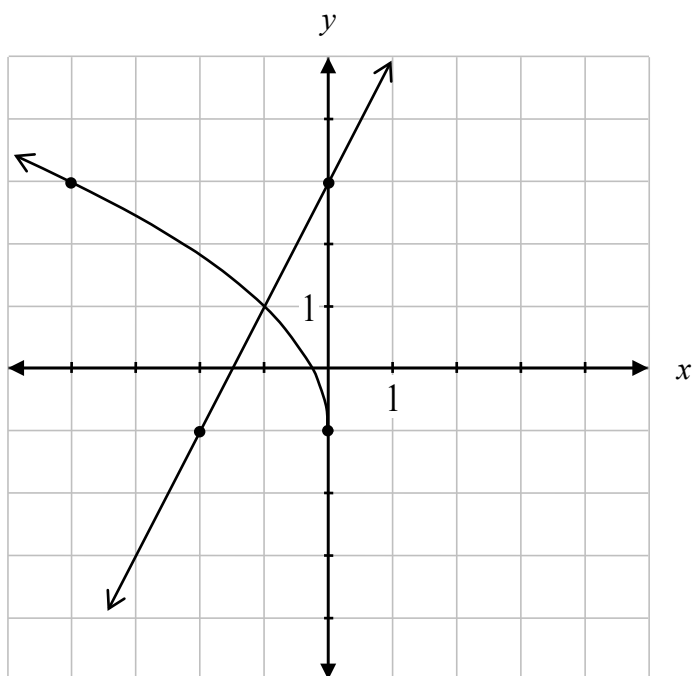
Describe her error.

Solution

When using the formula $s = \theta r$, the angle must be in radians. Ariane did not convert the central angle from degrees to radians.

1 mark

Using the graphs below, state the solution of the equation $2x + 3 = 2\sqrt{-x} - 1$.



Solution

$$x = -1$$

1 mark

Solve, algebraically.

$$\log_2(x+3) = 5 - \log_2(x-1)$$

Solution

Method 1

$$\log_2(x+3) + \log_2(x-1) = 5$$

$$\log_2[(x+3)(x-1)] = 5$$

1 mark for product law

$$x^2 + 2x - 3 = 2^5$$

1 mark for exponential form

$$x^2 + 2x - 3 = 32$$

$$x^2 + 2x - 35 = 0$$

$$(x+7)(x-5) = 0$$

$$\cancel{x = -7} \quad x = 5$$

½ mark for solving for the permissible value of x

½ mark for showing the rejection of the extraneous root

3 marks

Method 2

$$\log_2(x+3) + \log_2(x-1) = 5$$

$$\log_2[(x+3)(x-1)] = \log_2 2^5$$

1 mark for product law

$$x^2 + 2x - 3 = 32$$

1 mark for equating arguments

$$x^2 + 2x - 35 = 0$$

$$(x+7)(x-5) = 0$$

$$\cancel{x = -7} \quad x = 5$$

½ mark for the permissible value of x

½ mark for showing the rejection of the extraneous root

3 marks

Explain why the value of n must be greater than or equal to the value of r , when using ${}_nC_r$.

Solution

The number of objects to select, r , cannot be greater than the total number of objects, n .

1 mark

Question 9

R2, R3, R5

Given that $y = |x|$, determine the equation of the resulting function, $g(x)$, after the following transformations:

- reflection in the x -axis
- vertical translation 5 units down
- horizontal stretch by a factor of 3

Solution

$$g(x) = -\left|\frac{1}{3}x\right| - 5$$

1 mark for vertical reflection
1 mark for vertical translation
1 mark for horizontal stretch

3 marks

Explain why the graph of $y = \frac{x-1}{x^2+x-6}$ has a horizontal asymptote at $y = 0$.

Solution

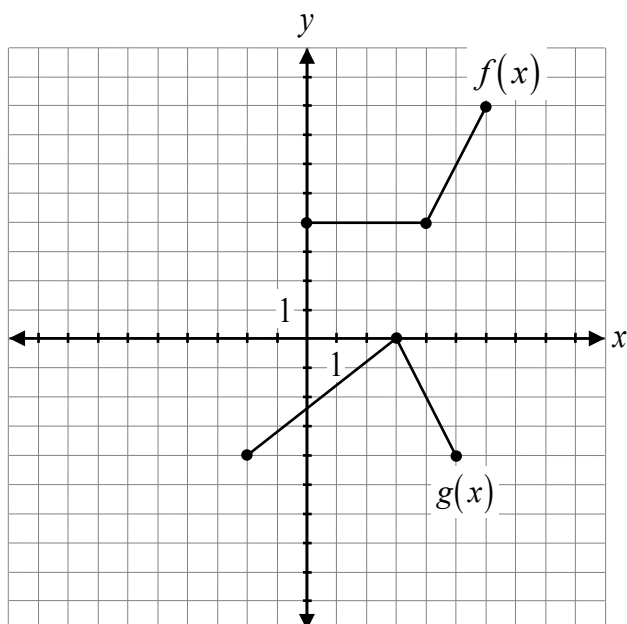
As x approaches positive or negative infinity, y approaches zero.

1 mark

or

The degree of the numerator is less than the degree of the denominator.

Given the graphs of $f(x)$ and $g(x)$, evaluate $g(f(2))$.

**Solution**

$$f(2) = 4 \quad \frac{1}{2} \text{ mark for the value of } f(2)$$

$$g(4) = -2 \quad \frac{1}{2} \text{ mark for consistent value of } g(f(2))$$

1 mark

Kennedy was asked to solve the equation $\tan \theta = 1$ over all real numbers.

Below is Kennedy's solution:

$$\tan \theta = 1$$
$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

Describe her error.

Solution

Kennedy did not include the general solution in her answer.

1 mark

Question 13

P2, P3

Solve, algebraically.

$${}_nC_3 = 3({}_nP_2)$$

Solution

$$\frac{n!}{(n-3)!3!} = \frac{3n!}{(n-2)!}$$

1 mark for substitution into equation (½ mark for each side)

$$\frac{n!(n-2)!}{(n-3)!3!} = 3n!$$

$$\frac{n!(n-2)\cancel{(n-3)!}}{\cancel{(n-3)!}} = 3!(3n!)$$

1 mark for factorial expansion

$$n-2 = 3!(3)$$

½ mark for simplification of factorials

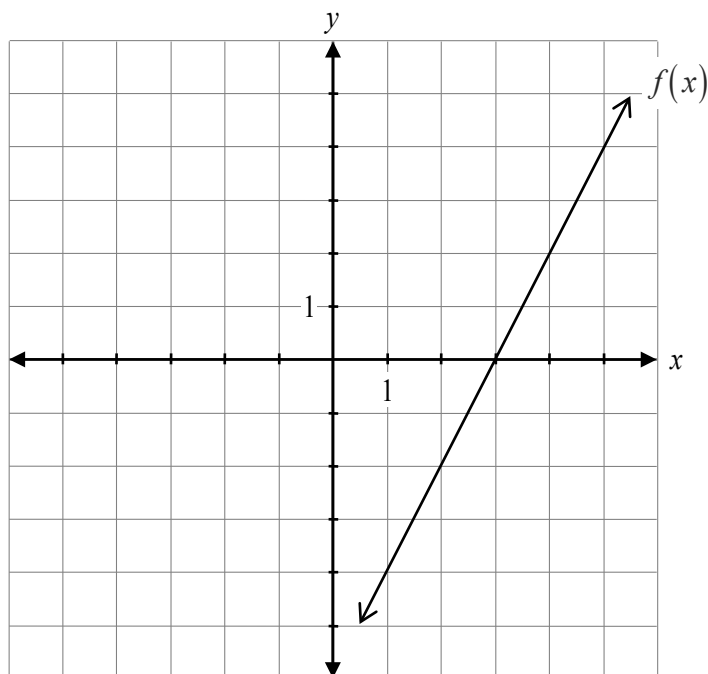
$$n-2 = 18$$

$$n = 20$$

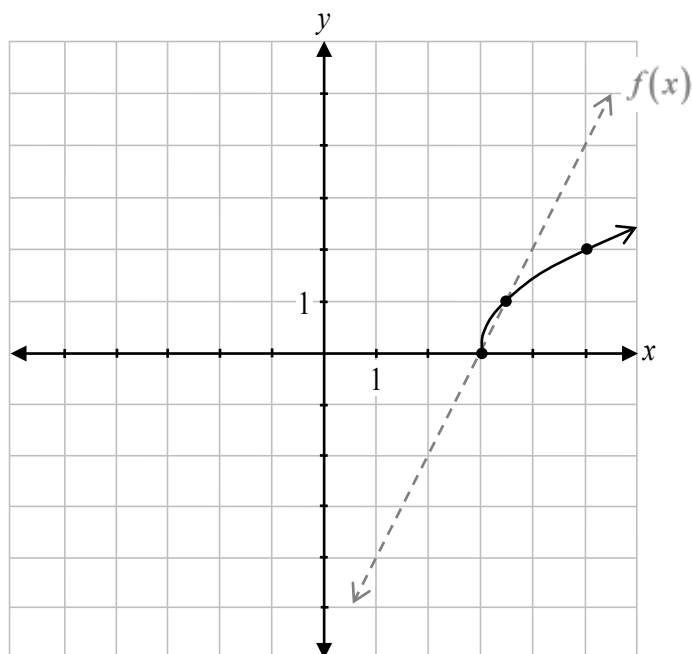
½ mark for solving for n

3 marks

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



Solution



1 mark for restricting domain
 $\frac{1}{2}$ mark for shape between invariant points
 $\frac{1}{2}$ mark for shape to the right of invariant points

2 marks

Prove the identity for all permissible values of θ .

$$\frac{\sec \theta - \tan \theta \sin \theta}{\tan \theta \sin \theta} = \csc^2 \theta - 1$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{\frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta} \cdot \sin \theta}{\frac{\sin \theta}{\cos \theta} \cdot \sin \theta}$	$\csc^2 \theta - 1$
$\frac{\frac{1 - \sin^2 \theta}{\cancel{\cos \theta}}}{\frac{\sin^2 \theta}{\cancel{\cos \theta}}}$	
$\frac{1 - \sin^2 \theta}{\sin^2 \theta}$	
$\frac{1}{\sin^2 \theta} - \frac{\sin^2 \theta}{\sin^2 \theta}$	
$\csc^2 \theta - 1$	

1 mark for substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Method 2

Left-Hand Side	Right-Hand Side
$\frac{\sec \theta}{\tan \theta \sin \theta} - 1$	$\csc^2 \theta - 1$
$\frac{\frac{1}{\cos \theta}}{\frac{\sin \theta}{\cos \theta}} - 1$	
$\frac{1}{\cos \theta} \cdot \frac{\cos \theta}{\sin^2 \theta} - 1$	
$\frac{1}{\sin^2 \theta} - 1$	
$\csc^2 \theta - 1$	

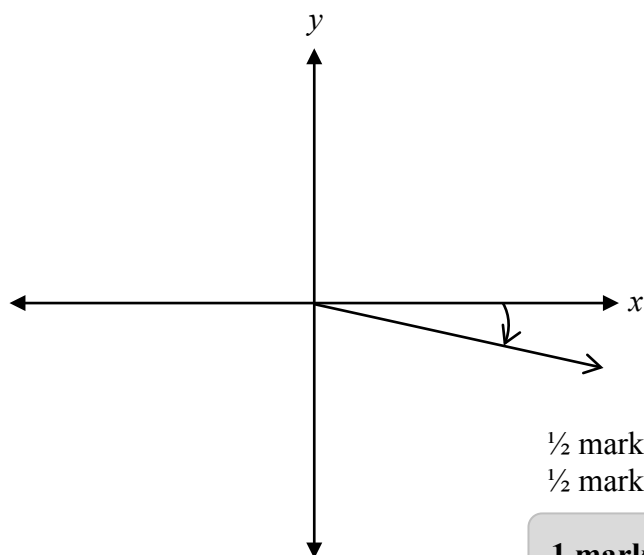
1 mark for substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Sketch the angle of $-\frac{\pi}{12}$ radians in standard position.

Solution

$\frac{1}{2}$ mark for an appropriate angle in quadrant IV
 $\frac{1}{2}$ mark for correct direction

1 mark

Note:

- If the directional arrow is not indicated, deduct an E1 error (final answer not stated)

Given that $h(x) = 2x^2 - 7x - 15$, determine possible equations of the functions $f(x)$ and $g(x)$ if $h(x) = f(x) \cdot g(x)$.

Solution

$$f(x) = \underline{2x + 3}$$

$$g(x) = \underline{x - 5}$$

1 mark for two correct factors of $h(x)$

1 mark

Note:

- Other answers are possible.

Question 18

R3

The range of $y = f(x)$ is $-6 \leq y \leq 12$. The range of the transformed function $y = af(x)$ is $-2 \leq y \leq 4$. Identify the value of a .

a) -3

b) $-\frac{1}{3}$

c) $\frac{1}{3}$

d) 3

Question 19

R8

Identify the expression which is equivalent to $3 \log y - \frac{1}{2} \log x + \log z$.

a) $\log \left(\frac{y^3}{\sqrt{xz}} \right)$

b) $\log \left(\frac{y^3 z}{\sqrt{x}} \right)$

c) $\log \left(\frac{y^3}{x^2 z} \right)$

d) $\log \left(\frac{y^3 z}{x^2} \right)$

Question 20

T1

Identify the measure of the angle $-\frac{2\pi}{9}$ in degrees.

a) -400°

b) -40°

c) 40°

d) 320°

Question 21

R6

If $y = f(x)$ has a domain of $[2, 5]$ and a range of $[6, 10]$, identify the domain of $y = f^{-1}(x)$.

a) $\left[\frac{1}{2}, \frac{1}{5}\right]$

b) $[-5, -2]$

c) $[-10, -6]$

d) $[6, 10]$

Question 22

R12

Identify which of the following is a polynomial function.

a) $p(x) = -\frac{1}{2}(x+2)^3(x-3)$

b) $p(x) = 2x^{\frac{1}{2}} + x - 3$

c) $p(x) = 3x^{-4} + x^2 - 6$

d) $p(x) = 2^x + 3$

Question 23

P4

Identify the total number of terms in the expansion of $(x - y)^9$.

a) 8

b) 9

c) 10

d) 11

Question 24

T6

Identify the exact value of $2 \cos^2(15^\circ) - 1$.

a) 1

b) $\frac{1}{2}$

c) $\frac{\sqrt{3}}{2}$

d) $\sqrt{3}$

Question 25

R2

The zeros of the function $y = f(x)$ are $x = -2$ and $x = 3$. Identify the zeros of the function $g(x) = 2f(x - 4)$.

a) $x = -6$ and $x = -1$

b) $x = 2$ and $x = 7$

c) $x = -4$ and $x = 6$

d) $x = 0$ and $x = 10$

Question 26

R7

Identify the value of $\log_4\left(\frac{1}{4}\right)$.

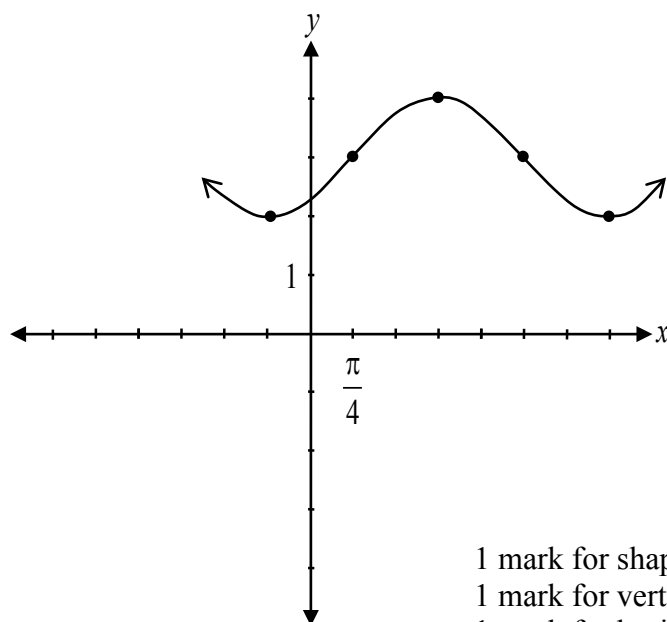
a) -16

b) -1

c) 1

d) 16

Sketch the graph of at least one period of the function $y = -\cos\left(x + \frac{\pi}{4}\right) + 3$.

Solution

- 1 mark for shape of a sinusoidal function with correct period
- 1 mark for vertical reflection
- 1 mark for horizontal translation
- 1 mark for vertical translation

4 marks

Justify that $(x - 5)$ is not a possible factor of the function $P(x) = x^3 - 3x^2 - 4x + 12$.

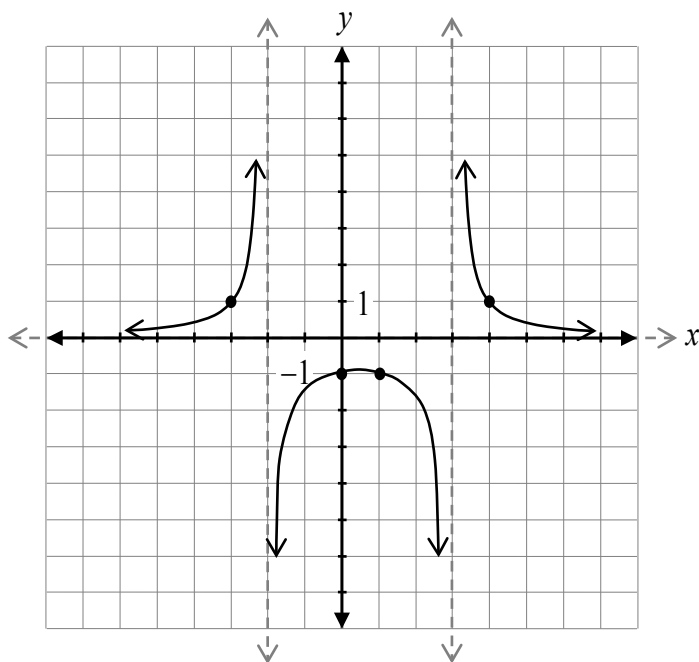
Solution

When $x = 5$ is substituted into $P(x)$, $P(5)$ does not equal 0.

1 mark

Sketch the graph of $f(x) = \frac{6}{(x+2)(x-3)}$ and state the y -intercept.

Solution



y -intercept: -1

1 mark for vertical asymptotic behaviour

($\frac{1}{2}$ mark for behaviour approaching $x = -2$; $\frac{1}{2}$ mark for behaviour approaching $x = 3$)

1 mark for horizontal asymptotic behaviour approaching $y = 0$

$1\frac{1}{2}$ marks for shape ($\frac{1}{2}$ mark for shape in each section)

$\frac{1}{2}$ mark for y -intercept

4 marks

Question 30

P2

Determine how many 3-digit odd numbers less than 300 are possible using the digits 1, 2, 3, 4, 5, 6 if repetition is not allowed.

Solution

$$\text{case 1: } \frac{1}{1} \cdot \frac{4}{-} \cdot \frac{2}{\text{odd}} = 8$$

$\frac{1}{2}$ mark for case 1

$$\text{case 2: } \frac{1}{2} \cdot \frac{4}{-} \cdot \frac{3}{\text{odd}} = 12$$

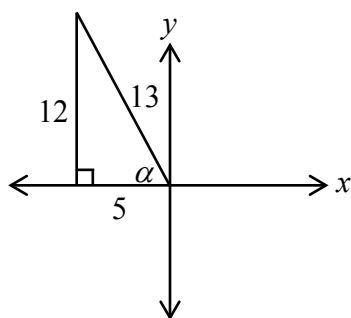
$\frac{1}{2}$ mark for case 2

$$8 + 12 = 20 \text{ numbers}$$

1 mark for addition of cases

2 marks

Given that $\cos \alpha = -\frac{5}{13}$ and $\sin \beta = \frac{2}{3}$, where α and β terminate in the same quadrant, determine the exact value of $\cos(\alpha - \beta)$.

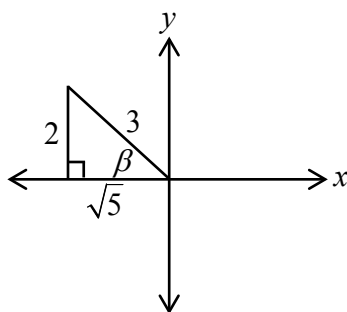
Solution

$$x^2 + y^2 = r^2$$

$$25 + y^2 = 169$$

$$y^2 = 144$$

$$y = \pm 12$$



$$x^2 + y^2 = r^2$$

$$x^2 + 4 = 9$$

$$x^2 = 5$$

$$x = \pm\sqrt{5}$$

$\frac{1}{2}$ mark for value of x
 $\frac{1}{2}$ mark for value of y

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$= \left(-\frac{5}{13}\right)\left(-\frac{\sqrt{5}}{3}\right) + \left(\frac{12}{13}\right)\left(\frac{2}{3}\right)$$

$$= \frac{5\sqrt{5}}{39} + \frac{24}{39}$$

$$= \frac{5\sqrt{5} + 24}{39}$$

$\frac{1}{2}$ mark for $\cos \beta$

$\frac{1}{2}$ mark for $\sin \alpha$

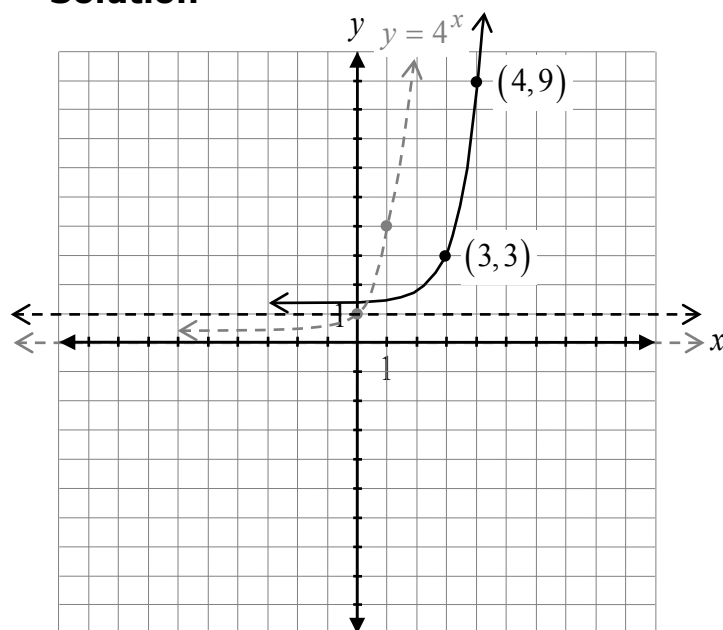
1 mark for substitution into correct identity

3 marks

Notes:

- Accept any of the following values for x : $x = \pm\sqrt{5}$, $x = \sqrt{5}$, or $x = -\sqrt{5}$.
- Accept any of the following values for y : $y = \pm 12$ or $y = 12$.

Given the graph of $y = 4^x$, sketch the graph of $y = 2(4)^{x-3} + 1$.

Solution

1 mark for vertical stretch
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Determine the coterminal angle of $\frac{\pi}{5}$ over the interval $[-2\pi, 0]$.

Solution

$$-\frac{9\pi}{5}$$

1 mark

State the domain of the graph of $y = \log(x - 4) - 8$.

Solution

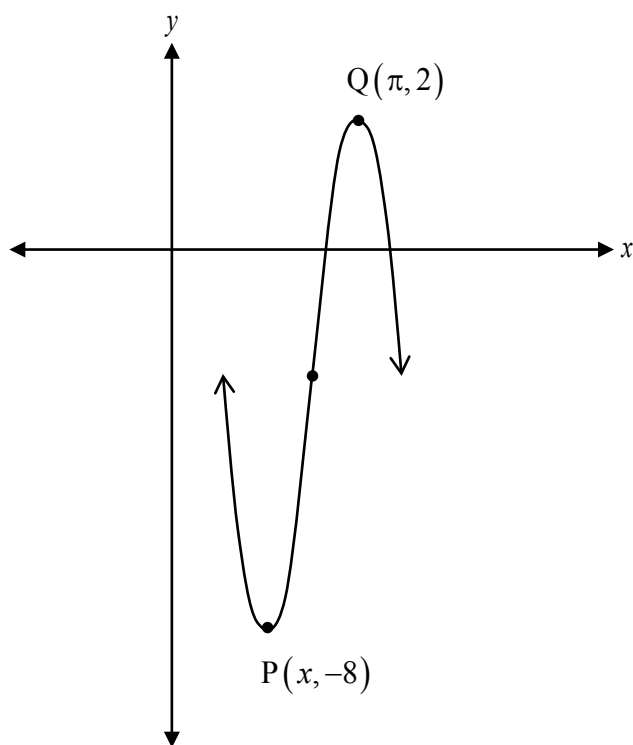
$$x > 4$$

or

$$(4, \infty)$$

1 mark

Given the graph of $y = 5 \sin \left[2 \left(x + \frac{\pi}{4} \right) \right] - 3$, determine the exact value of the x -coordinate in the point P.

**Solution**

$$x = \frac{\pi}{2}$$

1 mark

Verify that the following equation is true for $x = \frac{5\pi}{6}$.

$$\frac{\cos x}{1 - \sin x} = \frac{1 + \sin x}{\cos x}$$

Solution

Left-Hand Side	Right-Hand Side
$\frac{\cos \frac{5\pi}{6}}{1 - \sin \frac{5\pi}{6}}$	$\frac{1 + \sin \frac{5\pi}{6}}{\cos \frac{5\pi}{6}}$
$\frac{-\frac{\sqrt{3}}{2}}{1 - \frac{1}{2}}$	$\frac{1 + \frac{1}{2}}{-\frac{\sqrt{3}}{2}}$ 1 mark for exact values (½ mark for $\cos \frac{5\pi}{6}$; ½ mark for $\sin \frac{5\pi}{6}$)
$\frac{-\frac{\sqrt{3}}{2}}{\frac{1}{2}}$	$\frac{\frac{3}{2}}{-\frac{\sqrt{3}}{2}}$
$-\frac{\sqrt{3}}{1}$	$-\frac{3}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$ 1 mark for simplification (½ mark for LHS; ½ mark for RHS)
$-\sqrt{3}$	$-\sqrt{3}$

2 marks

Given that $(x + 1)$ is one of the factors of $P(x) = x^3 - x^2 + kx - 8$, determine the value of k .

Solution

Method 1

$$x = -1$$

$\frac{1}{2}$ mark for $x = -1$

$$0 = (-1)^3 - (-1)^2 + k(-1) - 8$$

1 mark for remainder theorem

$$0 = -1 - 1 - k - 8$$

$$k = -10$$

$\frac{1}{2}$ mark for solving for k

Method 2

$$x = -1$$

$\frac{1}{2}$ mark for $x = -1$

$$\begin{array}{r|rrrr} -1 & 1 & -1 & k & -8 \\ & & -1 & 2 & -k-2 \\ \hline & 1 & -2 & k+2 & -k-10 \end{array}$$

1 mark for synthetic division

$$-k - 10 = 0$$

$$k = -10$$

$\frac{1}{2}$ mark for solving for k

2 marks

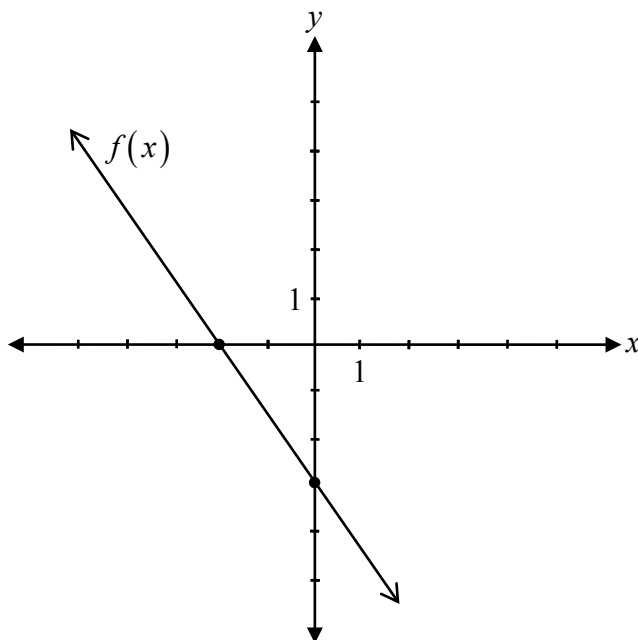
Given the function $f(x) = \sqrt{x}$, describe how to use transformations to determine the domain of the function $g(x) = f(x + 2) + 1$.

Solution

The graph of $g(x)$ is a horizontal translation 2 units to the left of the graph of $f(x)$, which changes the domain from $x \geq 0$ to $x \geq -2$.

1 mark

Given the graph of $y = f(x)$, state the equation of the vertical asymptote of $y = \frac{1}{f(x)}$.



Solution

$$x = -2$$

1 mark

Solve, algebraically.

$$16^x = 64^{2x-1}$$

Solution

$$4^{2x} = 4^{3(2x-1)}$$

1 mark for changing to a common base ($\frac{1}{2}$ mark for each)

$$4^{2x} = 4^{6x-3}$$

$\frac{1}{2}$ mark for exponent law

$$2x = 6x - 3$$

$\frac{1}{2}$ mark for equating exponents

$$3 = 4x$$

2 marks

$$\frac{3}{4} = x$$

Question 41

R11

Given that one of the factors of $P(x) = x^3 + 2x^2 - 5x - 6$ is $(x + 3)$, express $P(x)$ in completely factored form.

Solution

$$\begin{array}{r|rrrr} -3 & 1 & 2 & -5 & -6 \\ & \downarrow & -3 & 3 & 6 \\ \hline & 1 & -1 & -2 & 0 \end{array}$$

$\frac{1}{2}$ mark for $x = -3$

1 mark for synthetic division (or any equivalent strategy)

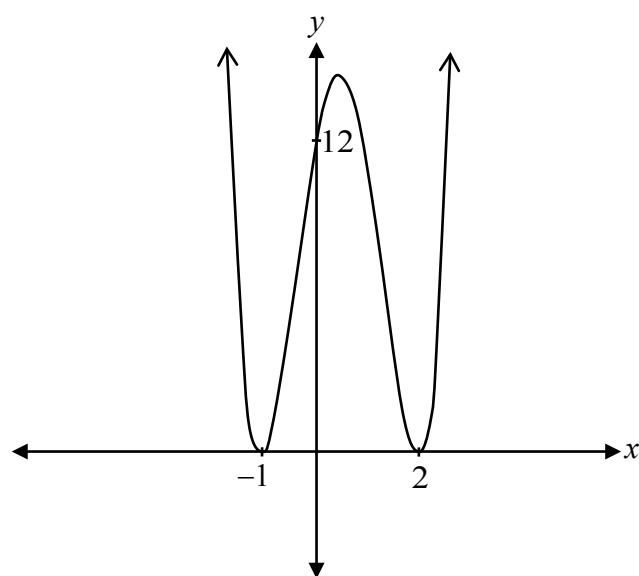
$$P(x) = (x + 3)(x^2 - x - 2)$$

$$P(x) = \underline{(x + 3)(x - 2)(x + 1)}$$

$\frac{1}{2}$ mark for consistent product of factors

2 marks

Sketch the graph of $p(x) = 3(x+1)^2(x-2)^2$.

Solution

1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

1 mark for multiplicity of 2 at $x = -1$ and at $x = 2$
($\frac{1}{2}$ mark for each)

$\frac{1}{2}$ mark for end behaviour

3 marks

Given that $f(x) = x^2 - 4$ and $g(x) = \sqrt{x}$, determine $f(g(x))$ and state its domain.

Solution

$$f(g(x)) = (\sqrt{x})^2 - 4$$

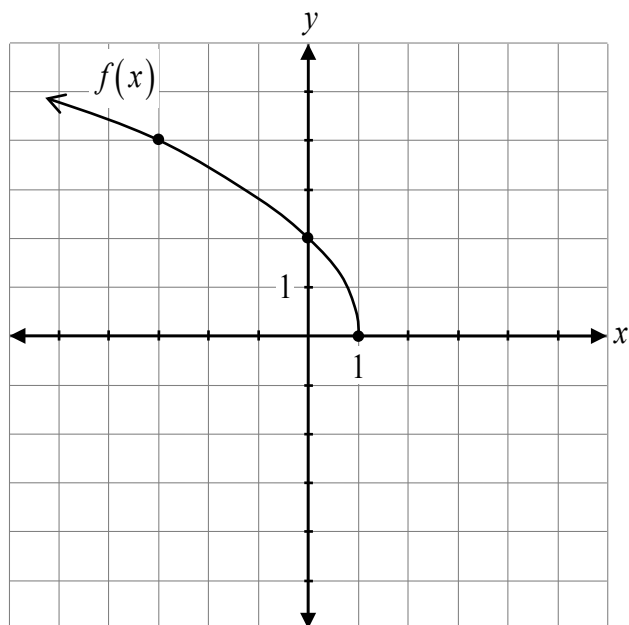
$$f(g(x)) = \underline{x - 4, x \geq 0}$$

1 mark for composite function

1 mark for domain consistent with composite function

2 marks

Determine a possible equation of the function $f(x)$.

**Solution**

$$f(x) = 2\sqrt{-(x-1)}$$

1 mark for vertical stretch
1 mark for horizontal reflection
1 mark for horizontal translation

3 marks

or

$$f(x) = \sqrt{-4(x-1)}$$

1 mark for horizontal compression
1 mark for horizontal reflection
1 mark for horizontal translation

3 marks

Explain why the graph of $y = \log_2 x$ does not have a y -intercept.

Solution

The domain of the graph is $x > 0$.

1 mark

or

There is a vertical asymptote at $x = 0$.

Evaluate.

$$\sin^2\left(-\frac{\pi}{3}\right) + \cos\left(\frac{17\pi}{6}\right)\sec\left(\frac{\pi}{6}\right)$$

Solution

$$\begin{aligned} \left(-\frac{\sqrt{3}}{2}\right)^2 + \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{2}{\sqrt{3}}\right) & \quad 1 \text{ mark for } \sin\left(-\frac{\pi}{3}\right) \text{ (}\frac{1}{2}\text{ mark for quadrant; } \frac{1}{2}\text{ mark for value)} \\ \frac{3}{4} - 1 & \quad 1 \text{ mark for } \cos\left(\frac{17\pi}{6}\right) \text{ (}\frac{1}{2}\text{ mark for quadrant; } \frac{1}{2}\text{ mark for value)} \\ -\frac{1}{4} & \quad 1 \text{ mark for } \sec\left(\frac{\pi}{6}\right) \text{ (}\frac{1}{2}\text{ mark for quadrant; } \frac{1}{2}\text{ mark for value)} \end{aligned}$$

3 marks

Determine the coordinates of the point of discontinuity (hole) on the graph of $y = \frac{x^2 - 3x}{x}$.

Solution

$(0, -3)$ 1 mark for point of discontinuity (hole) at $x = 0$

1 mark

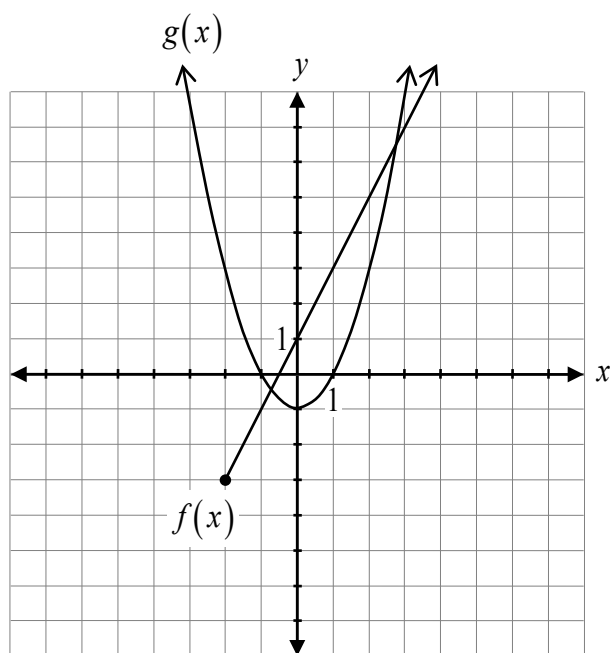
Note:

- Deduct $\frac{1}{2}$ mark for procedural error of incorrect y -value.

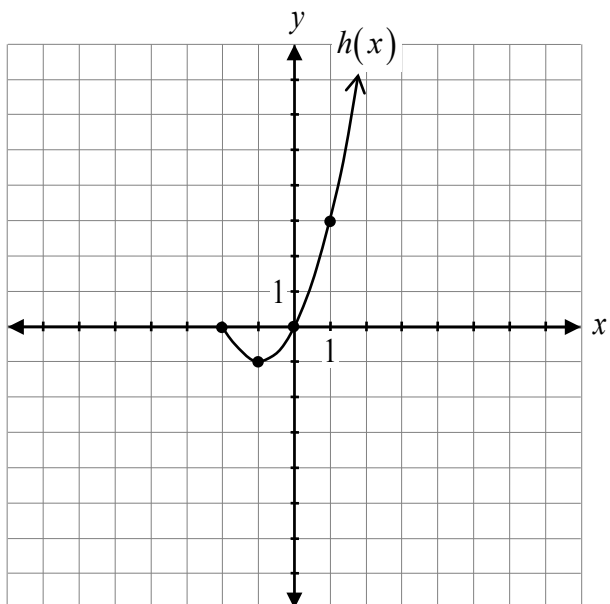
Question 48

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) + g(x)$.



Solution

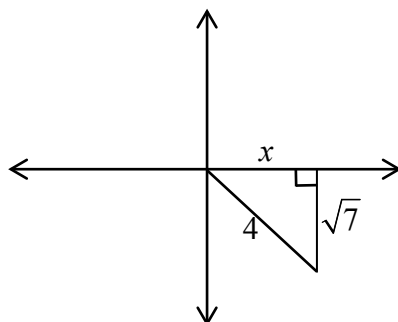


1 mark for operation of addition
1 mark for restricted domain

2 marks

Given that $\csc \theta = -\frac{4}{\sqrt{7}}$ and $\cos \theta > 0$, determine the exact value of $\tan \theta$.

Solution



$$\sin \theta = -\frac{\sqrt{7}}{4}$$

$$x^2 + (\sqrt{7})^2 = (4)^2$$

½ mark for substitution

$$x^2 = 16 - 7$$

$$x^2 = 9$$

$$x = \pm 3$$

½ mark for solving for x

$$\tan \theta = -\frac{\sqrt{7}}{3}$$

1 mark for $\tan \theta$ (½ mark for quadrant; ½ mark for value)

2 marks

Note:

- Accept any of the following values for x : $x = \pm 3$, $x = 3$, or $x = -3$.